

# DBMS LAB - 1 : Table Creation and Insertion

## 1.Student table

```
1 CREATE TABLE Student (  
2     Student_id INT PRIMARY KEY,  
3     Student_name VARCHAR(100) NOT NULL,  
4     Email VARCHAR(100) UNIQUE,  
5     Phone VARCHAR(15),  
6     DOB DATE,  
7     Gender VARCHAR(10),  
8     Department_id INT,  
9     Course VARCHAR(20),  
10    Cgpa INT(4)  
11 );  
12  
13 DESC Student;  
14  
15 INSERT INTO Student VALUES  
16 (1, 'Pratham', 'pratham123@gmail.com', '9876543210', '2007-04-09', 'Male', 101, 'BCA', 8.7),  
17 (2, 'Shreya', 'shreyan@gmail.com', '9876543211', '2007-08-18', 'Female', 102, 'BSC', 8.9),  
18 (3, 'Annie', 'annie@gmail.com', '9876543212', '2016-12-11', 'Female', 101, 'BSC', 8.3);  
19  
20 Select * from Student ;
```

Output:

| Field         | Type         | Null | Key | Default | Extra |
|---------------|--------------|------|-----|---------|-------|
| Student_id    | int          | NO   | PRI | NULL    |       |
| Student_name  | varchar(100) | NO   |     | NULL    |       |
| Email         | varchar(100) | YES  | UNI | NULL    |       |
| Phone         | varchar(15)  | YES  |     | NULL    |       |
| DOB           | date         | YES  |     | NULL    |       |
| Gender        | varchar(10)  | YES  |     | NULL    |       |
| Department_id | int          | YES  |     | NULL    |       |
| Course        | varchar(20)  | YES  |     | NULL    |       |
| Cgpa          | int          | YES  |     | NULL    |       |

  

| Student_id | Student_name | Email                | Phone      | DOB        | Gender | Department_id | Course | Cgpa |
|------------|--------------|----------------------|------------|------------|--------|---------------|--------|------|
| 1          | Pratham      | pratham123@gmail.com | 9876543210 | 2007-04-09 | Male   | 101           | BCA    | 9    |
| 2          | Shreya       | shreyan@gmail.com    | 9876543211 | 2007-08-18 | Female | 102           | BSC    | 9    |
| 3          | Annie        | annie@gmail.com      | 9876543212 | 2016-12-11 | Female | 101           | BSC    | 8    |

## 2.Faculty table

```
1 CREATE TABLE Faculty (  
2     Faculty_id INT PRIMARY KEY,  
3     Faculty_name VARCHAR(100) NOT NULL,  
4     Email VARCHAR(100) UNIQUE,  
5     Phone VARCHAR(15),  
6     Department_id INT  
7 );  
8  
9 DESC Faculty;  
10  
11 INSERT INTO Faculty VALUES  
12 (1, 'Dr. Ramesh Kumar', 'ramesh@gmail.com', '9876543210', 101),  
13 (2, 'Dr. Priya Sharma', 'priya@gmail.com', '9876543211', 102),  
14 (3, 'Dr. Anil Patel', 'anil@gmail.com', '9876543212', 103);  
15  
16 SELECT * FROM Faculty;
```

Output:

| Field         | Type         | Null | Key | Default | Extra |
|---------------|--------------|------|-----|---------|-------|
| Faculty_id    | int          | NO   | PRI | NULL    |       |
| Faculty_name  | varchar(100) | NO   |     | NULL    |       |
| Email         | varchar(100) | YES  | UNI | NULL    |       |
| Phone         | varchar(15)  | YES  |     | NULL    |       |
| Department_id | int          | YES  |     | NULL    |       |

  

| Faculty_id | Faculty_name     | Email            | Phone      | Department_id |
|------------|------------------|------------------|------------|---------------|
| 1          | Dr. Ramesh Kumar | ramesh@gmail.com | 9876543210 | 101           |
| 2          | Dr. Priya Sharma | priya@gmail.com  | 9876543211 | 102           |
| 3          | Dr. Anil Patel   | anil@gmail.com   | 9876543212 | 103           |

### 3.Course table

```
CREATE TABLE Course (  
    Course_id INT PRIMARY KEY,  
    Course_name VARCHAR(100) NOT NULL,  
    Credits INT NOT NULL,  
    Department_id INT  
);
```

```
DESC Course;
```

```
INSERT INTO Course VALUES  
(1, 'Database Management', 4, 101),  
(2, 'Java Programming', 3, 102),  
(3, 'Computer Networks', 4, 103);
```

```
SELECT * FROM Course;
```

Output:

| Field         | Type         | Null | Key | Default | Extra |
|---------------|--------------|------|-----|---------|-------|
| Course_id     | int          | NO   | PRI | NULL    |       |
| Course_name   | varchar(100) | NO   |     | NULL    |       |
| Credits       | int          | NO   |     | NULL    |       |
| Department_id | int          | YES  |     | NULL    |       |

  

| Course_id | Course_name         | Credits | Department_id |
|-----------|---------------------|---------|---------------|
| 1         | Database Management | 4       | 101           |
| 2         | Java Programming    | 3       | 102           |
| 3         | Computer Networks   | 4       | 103           |

## 4.Enrollment table

```
1 CREATE TABLE Enrollment (  
2     Enrollment_id INT PRIMARY KEY,  
3     Student_id INT NOT NULL,  
4     Course_id INT NOT NULL,  
5     Semester VARCHAR(20),  
6     Enrollment_date DATE  
7 );  
8  
9 DESC Enrollment;  
10  
11 INSERT INTO Enrollment VALUES  
12 (1001, 1, 188, 'Semester 1', '2026-01-10'),  
13 (1002, 2, 255, 'Semester 1', '2026-01-11'),  
14 (1003, 3, 335, 'Semester 2', '2026-06-15');  
15  
16 SELECT * FROM Enrollment;
```

Output:

| Field           | Type        | Null | Key | Default | Extra |
|-----------------|-------------|------|-----|---------|-------|
| Enrollment_id   | int         | NO   | PRI | NULL    |       |
| Student_id      | int         | NO   |     | NULL    |       |
| Course_id       | int         | NO   |     | NULL    |       |
| Semester        | varchar(20) | YES  |     | NULL    |       |
| Enrollment_date | date        | YES  |     | NULL    |       |

  

| Enrollment_id | Student_id | Course_id | Semester   | Enrollment_date |
|---------------|------------|-----------|------------|-----------------|
| 1001          | 1          | 188       | Semester 1 | 2026-01-10      |
| 1002          | 2          | 255       | Semester 1 | 2026-01-11      |
| 1003          | 3          | 335       | Semester 2 | 2026-06-15      |

## 5. Classroom table

```
1  CREATE TABLE Classroom (  
2      Classroom_id INT PRIMARY KEY,  
3      Room_number VARCHAR(20) NOT NULL,  
4      Building VARCHAR(100),  
5      Capacity INT  
6  );  
  
8  DESC Classroom;  
  
10 INSERT INTO Classroom  
11 VALUES  
12 (1, 'A101', 'A Block ', 600),  
13 (2, 'B202', 'B Block', 500),  
14 (3, 'C303', 'C Block', 400);  
  
16 SELECT * FROM Classroom;
```

Output:

| Field        | Type         | Null | Key | Default | Extra |
|--------------|--------------|------|-----|---------|-------|
| Classroom_id | int          | NO   | PRI | NULL    |       |
| Room_number  | varchar(20)  | NO   |     | NULL    |       |
| Building     | varchar(100) | YES  |     | NULL    |       |
| Capacity     | int          | YES  |     | NULL    |       |

  

| Classroom_id | Room_number | Building | Capacity |
|--------------|-------------|----------|----------|
| 1            | A101        | A Block  | 600      |
| 2            | B202        | B Block  | 500      |
| 3            | C303        | C Block  | 400      |

## 6.Exam table

```
CREATE TABLE Exam (  
    Exam_id INT PRIMARY KEY,  
    Course_id INT NOT NULL,  
    Exam_name VARCHAR(50),  
    Exam_date DATE,  
    Total_marks INT  
);  
  
DESC Exam;  
  
INSERT INTO Exam  
VALUES  
(1, 1234, 'CIE-1', '2026-08-10', 20),  
(2, 2658, 'CIE-2', '2026-08-12', 25),  
(3, 3574, 'CIE-3', '2026-08-15', 25),  
(4, 4456, 'SEE ', '2026-12-2 ', 30 );  
  
SELECT * FROM Exam;
```

Output:

| Field       | Type        | Null | Key | Default | Extra |
|-------------|-------------|------|-----|---------|-------|
| Exam_id     | int         | NO   | PRI | NULL    |       |
| Course_id   | int         | NO   |     | NULL    |       |
| Exam_name   | varchar(50) | YES  |     | NULL    |       |
| Exam_date   | date        | YES  |     | NULL    |       |
| Total_marks | int         | YES  |     | NULL    |       |

  

| Exam_id | Course_id | Exam_name | Exam_date  | Total_marks |
|---------|-----------|-----------|------------|-------------|
| 1       | 1234      | CIE-1     | 2026-08-10 | 20          |
| 2       | 2658      | CIE-2     | 2026-08-12 | 25          |
| 3       | 3574      | CIE-3     | 2026-08-15 | 25          |
| 4       | 4456      | SEE       | 2026-12-02 | 30          |

## 7.Fee payment table

```
1 CREATE TABLE Fee_Payment (  
2     Payment_id INT PRIMARY KEY,  
3     Student_id INT NOT NULL,  
4     Amount DECIMAL(10,2),  
5     Payment_date DATE,  
6     Payment_status VARCHAR(20)  
7 );  
  
8  
9 DESC Fee_Payment;  
  
10  
11 INSERT INTO Fee_Payment  
12 VALUES  
13 (1, 1423, 25000.00, '2026-07-01', 'Paid'),  
14 (2, 2234, 30000.00, '2026-07-05', 'Paid'),  
15 (3, 3265, 28000.00, '2026-07-10', 'Pending');  
16  
17 SELECT * FROM Fee_Payment;
```

Output:

| Field          | Type          | Null | Key | Default | Extra |
|----------------|---------------|------|-----|---------|-------|
| Payment_id     | int           | NO   | PRI | NULL    |       |
| Student_id     | int           | NO   |     | NULL    |       |
| Amount         | decimal(10,2) | YES  |     | NULL    |       |
| Payment_date   | date          | YES  |     | NULL    |       |
| Payment_status | varchar(20)   | YES  |     | NULL    |       |

  

| Payment_id | Student_id | Amount   | Payment_date | Payment_status |
|------------|------------|----------|--------------|----------------|
| 1          | 1423       | 25000.00 | 2026-07-01   | Paid           |
| 2          | 2234       | 30000.00 | 2026-07-05   | Paid           |
| 3          | 3265       | 28000.00 | 2026-07-10   | Pending        |

## DDL COMMANDS

```
CREATE TABLE Student (  
    Student_id INT PRIMARY KEY,  
    Student_name VARCHAR(100) NOT NULL,  
    Email VARCHAR(100) UNIQUE,  
    Phone VARCHAR(15),  
    DOB DATE,  
    Gender VARCHAR(10),  
    Department_id INT,  
    Course VARCHAR(20),  
    Cgpa INT  
);
```

```
DESC Student;
```

```
ALTER TABLE Student  
ADD Address VARCHAR(100);
```

```
DESC Student;
```

```
DROP TABLE Student;
```

Output:

| Field         | Type         | Null | Key | Default | Extra |
|---------------|--------------|------|-----|---------|-------|
| Student_id    | int          | NO   | PRI | NULL    |       |
| Student_name  | varchar(100) | NO   |     | NULL    |       |
| Email         | varchar(100) | YES  | UNI | NULL    |       |
| Phone         | varchar(15)  | YES  |     | NULL    |       |
| DOB           | date         | YES  |     | NULL    |       |
| Gender        | varchar(10)  | YES  |     | NULL    |       |
| Department_id | int          | YES  |     | NULL    |       |
| Course        | varchar(20)  | YES  |     | NULL    |       |
| Cgpa          | int          | YES  |     | NULL    |       |
| Field         | Type         | Null | Key | Default | Extra |
| Student_id    | int          | NO   | PRI | NULL    |       |
| Student_name  | varchar(100) | NO   |     | NULL    |       |
| Email         | varchar(100) | YES  | UNI | NULL    |       |
| Phone         | varchar(15)  | YES  |     | NULL    |       |
| DOB           | date         | YES  |     | NULL    |       |
| Gender        | varchar(10)  | YES  |     | NULL    |       |
| Department_id | int          | YES  |     | NULL    |       |
| Course        | varchar(20)  | YES  |     | NULL    |       |
| Cgpa          | int          | YES  |     | NULL    |       |
| Address       | varchar(100) | YES  |     | NULL    |       |